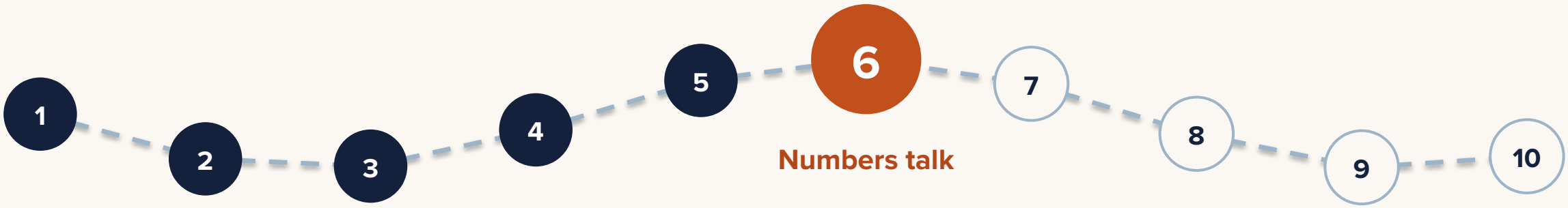
PYTHON FOR SUPPLY CHAIN

Session 6 · Thu 3 Dec 2026

# Numbers talk

merge, bar charts, line charts, finding titles




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# Today

1

## Merge

`count rows before, after`

2

## Bar chart

`plt.barh`

3

## Line chart

`plt.plot`

4

## Finding titles

`say what to see`

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# merge is your VLOOKUP




```
product_value = product_value.merge(product_names, on="stock_code", how="left")
```

on = the key both tables share

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# Count rows before and after

**3,066**
rows before



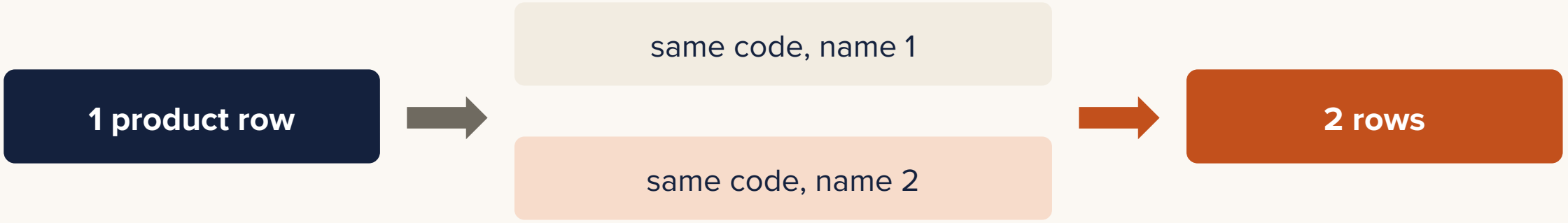
`merge`

**3,107**
rows after

41 extra rows and no error message.

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# A repeated key makes the table grow




**40 codes have two names → 41 extra rows**

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# Keep one name per code

```
product_names = sales_clean[["stock_code", "description"]]
product_names = product_names.drop_duplicates(subset="stock_code")
```

| 3,066 | = | 3,066 |
| --- | --- | --- |
| before | | after |

Same number: the merge is safe.

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# A chart, step by step

1. **`plt.figure()`** — a canvas
2. **`plt.barh()`** — the bars
3. **`plt.title()`** — the finding
4. **`plt.xlabel()`** — the units
5. **`plt.tight_layout()`** — tidy
6. **`plt.show()`** — show it

Every chart needs a title and axis labels.

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# The title says the finding

**Bar chart of revenue**

no

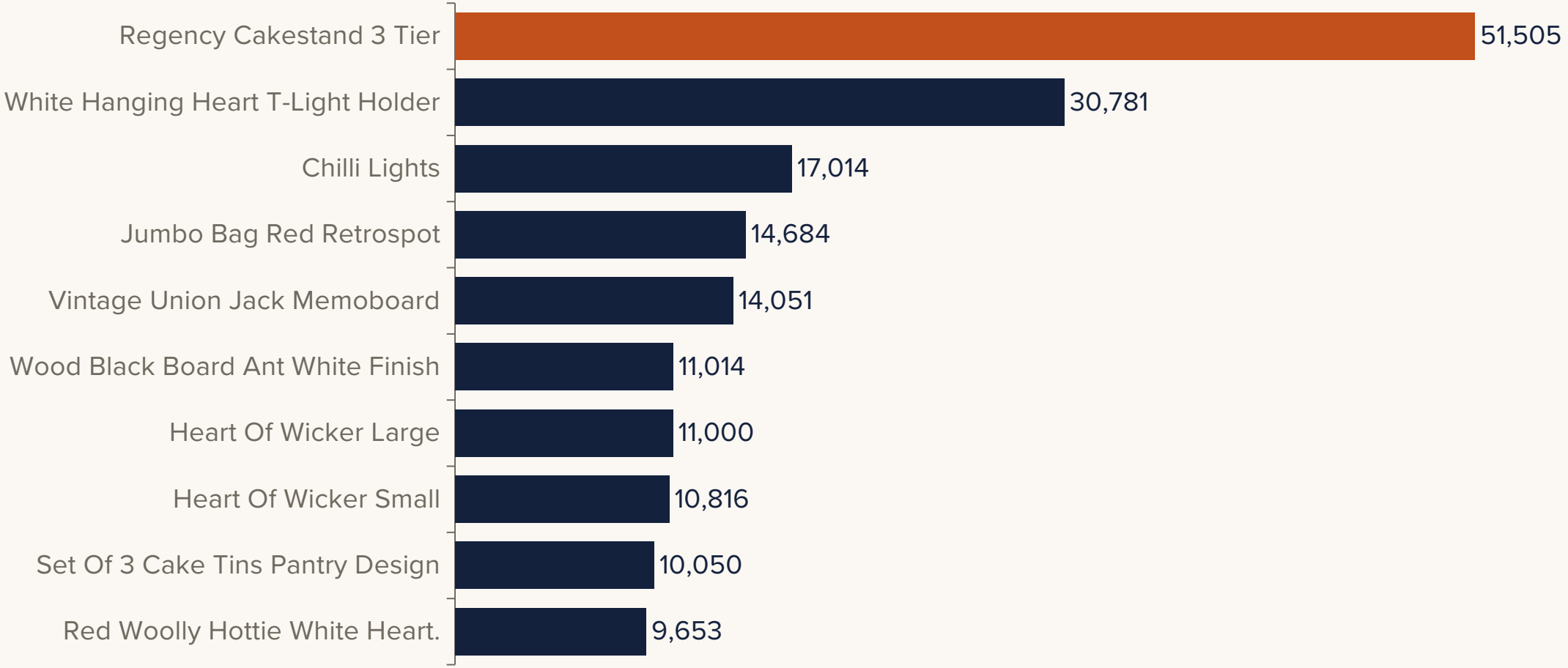
**The top 10 products are only 9.6% of revenue**

yes

A manager gets it in five seconds.

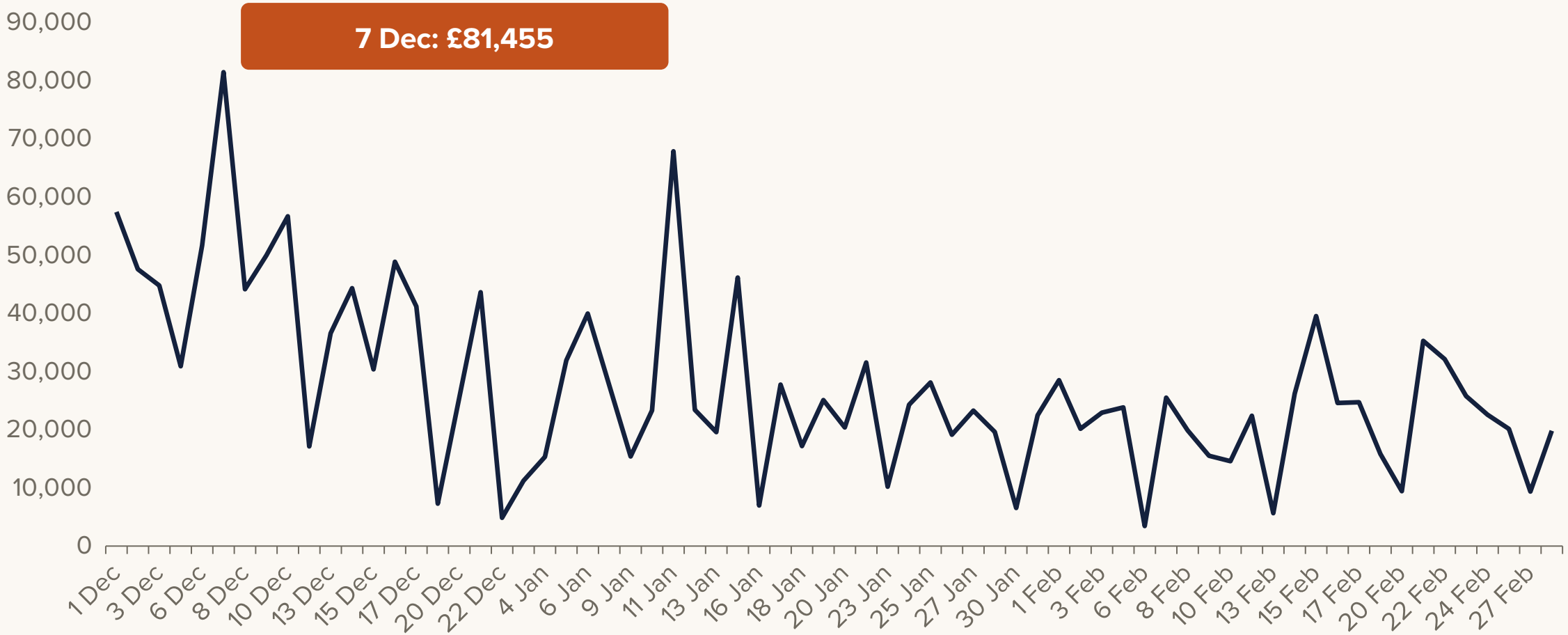
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# The top 10 products are only 9.6% of revenue




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# Revenue peaked in the first week of December




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# Three chart mistakes

### No title, no labels

the reader guesses

```
plt.title()  plt.xlabel()
```

### Labels on top of each other

names too long

```
plt.barh()  rotation=45
```

### Too many messages

the eye gets lost

```
one chart, one message
```

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# seaborn: a chart template

```
import seaborn as sns

sns.barplot(data=top_countries, x="revenue", y="country")
plt.title("Outside the UK, the Netherlands and Ireland lead")
plt.xlabel("Revenue (pounds)")
plt.show()
```

Same titles and labels, nicer defaults

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# Case hunt: due Session 8

1. **The problem**
2. **Who is affected**
3. **What data would show it**
4. **One question for data**
5. **Python, SQL or a dashboard?**

One page. Two minutes in Session 8.

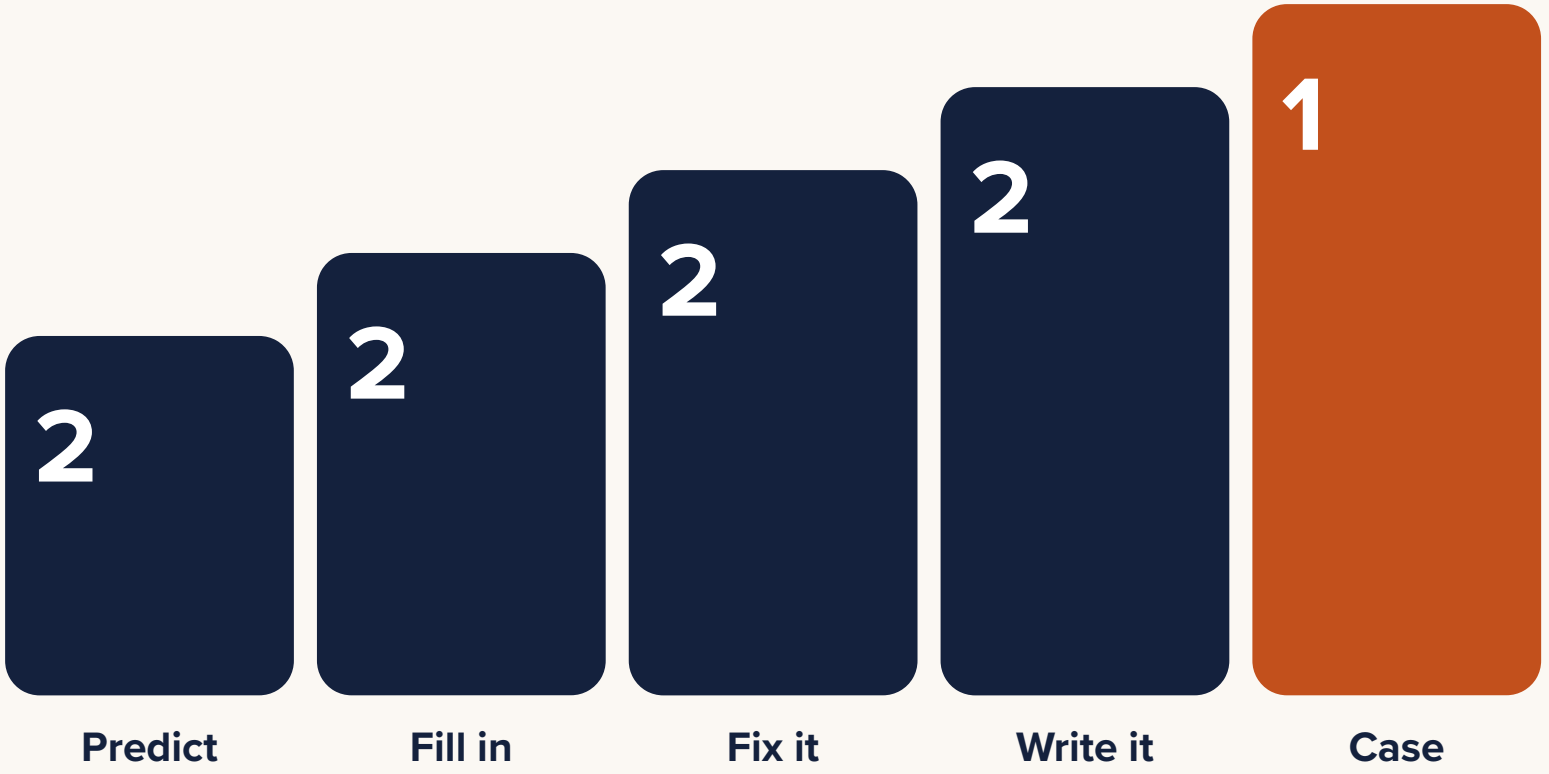
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# Your chart toolkit

| Code | Does |
|---|---|
| `a.merge(b, on="key", how="left")` | VLOOKUP: add columns |
| `len(a) before and after` | check the merge |
| `.drop_duplicates(subset="key")` | one row per key |
| `plt.barh(names, values)` | a bar chart |
| `plt.plot(days, values)` | a line chart |
| `plt.title()  plt.xlabel()` | the finding, the units |
| `.dt.date` | the day from a date |

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# Now in Colab




**Notebook**

s6_numbers_talk

**Homework**

seaborn, case hunt, Power BI

**Recall**

Fri 4 to Sun 6

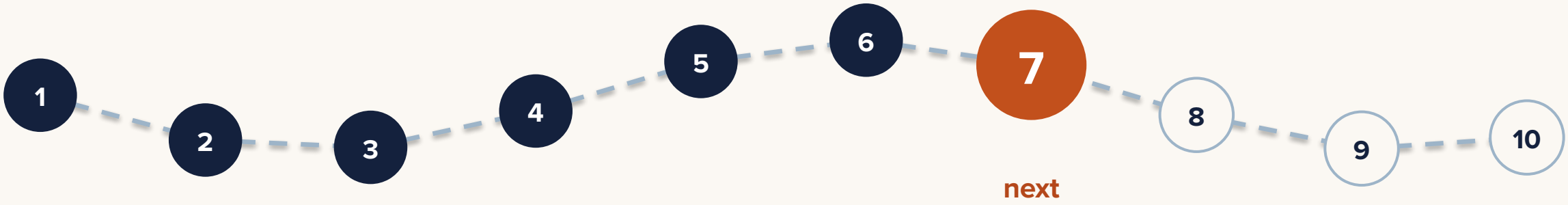
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# Thank you

**Next: Session 7**

## What will your project ask?

Mon 7 Dec




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